

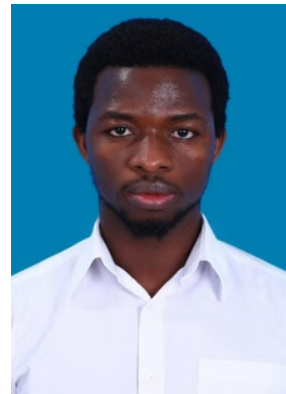
ALOUTE SANA (阿鲁特)

☎ +86 131 4139 2484 ✉ elsone121617@outlook.com

📍 Beijing City, China 🌐 Nationality : Burkina Faso

🌐 Github : <https://github.com/sanaaloute>

🗣 Speak : English - French - Chinese



PROFESSIONAL EXPERIENCES

Senior AI Fullstack Engineer

Shenzhen Depei Open Source Data Technology Co., Ltd | January 2026 – Present

- Architected an AI code generation platform using LangGraph and Pydantic to produce production-ready websites from natural language prompts
- Designed end-to-end system architecture spanning frontend, backend, and AI pipelines for scalability and reliability
- Built frontend with Next.js/React and backend with Supabase (auth, database, real-time features)
- Implemented interactive AI editing allowing users to modify generated UI components via natural language
- Enabled one-click GitHub push, Supabase integration, and Vercel deployment from the platform
- Established CI/CD pipelines for automated testing and seamless multi-environment deployments
- Led technical decisions on AI orchestration and platform scalability, aligning engineering with product strategy
- Built an ecommerce website to sell AI conversation devices, called daacoo.

Websites : www.daacoo.com, www.lovecode.com, www.dpqq.com, www.gitcc.com

AI/ML Engineer

Algento (北京奥金拓网络科技有限公司) | Beijing, China | June 2024 – December 2025

- Built RAG-powered employee service chatbots using LLMs, improving response relevance and user satisfaction
- Deployed LLM guardrails reducing safety violations (violence, sexual content, hate speech) by 80%
- Developed FastAPI-based voice-to-text chatbot with audio transcription, reducing user input time and increasing engagement by 20%
- Created agentic image generation/editing pipelines (Qwen-Image, Nano Banana, Kolors) for ad banner production, reducing manual design time by 90%
- Built a nationality prediction model with 98% accuracy, improving recommendation system performance by 60%

Algorithm Engineer | Internship

SFEM (北京钢流电磁技术有限公司) | Beijing, China | January 2024 – June 2024

- Developed a time series forecasting model for wind turbine power prediction with 80% forecast precision
- Led data collection, preprocessing, and feature engineering to optimize model accuracy
- Conducted model validation and performance evaluation to ensure production reliability

Algorithm Engineer | Internship

VMware | Beijing, China | October 2023 – January 2024

- Designed and developed page recognition algorithms for end-to-end user experiences
- Collaborated with cross-functional design teams throughout the full development lifecycle
- Contributed to company-wide graduate program initiatives and technical projects

PERSONAL PROJECTS

AI Web Builder | www.ai-web-builder.com

- AI-powered platform that generates full-stack websites from natural language prompts using multi-agent orchestration
- Designed and implemented a LangGraph/LangChain-based agentic workflow with 20+ specialized agents (planner, designer, debugger, executor, reviewer) for automated website generation
- Built an end-to-end system where user prompts trigger agent orchestration, dynamically routing tasks to the most appropriate agents based on intent detection
- Integrated visual component editing allowing users to select UI elements in-browser and modify them via natural language commands
- Implemented multi-export capabilities including Supabase persistence, GitHub push, and Coolify deployment, plus ZIP downloads for code portability
- Developed AI-driven debugging where users can request automated error fixes directly through the interface

Tech Stack: **LangGraph, LangChain, Python, Supabase, React/Next.js, Coolify**

BRVM RealTimeStock AI Agent | @BrvmBot (telegram)

- AI-driven Telegram bot for monitoring and analyzing BRVM stock market portfolios with multi-agent intelligence
- Architected a LangGraph-based agentic system featuring 1 master agent orchestrating 6 specialized subagents for distinct financial analysis tasks
- Enabled investors to perform complex portfolio analytics including:
 - Stock price plotting across custom date ranges
 - P&L calculations per stock and across entire portfolios
 - Multi-stock price comparisons for informed investment decisions

- Implemented Chroma vector database for efficient static data storage and SQLite for structured portfolio data management
- Deployed as a Telegram bot (@BrvmBot) providing real-time market intelligence to investors
- Designed conversation flows enabling users to create, update, read, and delete stock information via natural language interactions

Tech Stack: **LangGraph, LangChain, Python, ChromaDB, SQLite, Telegram Bot API**

Barkosem | www.barkosem.com

- Cross-platform e-commerce marketplace connecting Chinese suppliers with African B2B buyers
- Built a full-stack marketplace ecosystem consisting of:
 - Cross-platform mobile apps (Flutter - Android/iOS, awaiting store approval)
 - Vendor management panel and admin console
 - Landing page and customer-facing web interface (Next.js)
- Designed a microservices architecture with a NestJS API Gateway orchestrating backend services
- Implemented Supabase as the primary database solution for real-time data synchronization across platforms
- Created a multi-user ecosystem enabling bulk product transactions between international suppliers and African customers

Tech Stack: **Next.js, Flutter, NestJS, Supabase, REST APIs, Microservices Architecture**

EDUCATION

MASTER DEGREE IN COMPUTER SCIENCE AND TECHNOLOGY | 2021 ~ 2024

- **Beijing University of Aeronautics and Astronautics**
- Research Topic : Person re-identification in panoramic view based on Transformers.

BACHELOR IN ELECTRONIC INFORMATION ENGINEERING | 2018 ~ 2021

- **Beijing University of Aeronautics and Astronautics**
- Graduation project : Artificial Intelligence algorithm for fast optical data sampling processing.

CERTIFICATION

- IBM - RAG and Agentic AI Capstone Project
- Google - Cloud AI Infrastructure
- IBM - Develop Generative AI Application

CORE SKILLS

- Machine Learning & AI: NLP, Computer Vision, LLMs, MCP, Agent
- Agent orchestration : LangGraph, CrewAI, BeeAI, Pydantic
- Languages: Python, TypeScript
- AI coding : Claude, Cursor, Kimi Code
- API framework : FastAPI, Nodejs + Express, Flask, NestJs
- Cloud & DevOps: Docker, Git, AWS
- Database : MongoDB, PostgreSQL, MySQL, Firebase, Supabase
- Soft Skills: Problem-solving, Team collaboration, Project management.